

P1) Book Problem 2.43: Without substantial work, the operation of OA3 is unknown as it does not clearly have negative feedback; however, you may assume that it still operates such that virtual short analysis is still applicable (which it is as negative feedback is provided by OA2 within the loop). Further, although the problem states the wiper of R1 is “nominally positioned halfway,” you may assume that it is indeed positioned at the halfway point. R1 is used to “trim” the circuit; accounting for component tolerances to ensure equal gain between both V1 and V2

2.43 Compared with the classical triple-op-amp IA, the IA of Fig. P2.43 (see *EDN*, Oct. 1, 1992, p. 115) uses fewer resistances. The wiper, nominally positioned halfway, is used to maximize the CMRR. Show that $v_O = (1 + 2R_2/R_1)(v_2 - v_1)$.

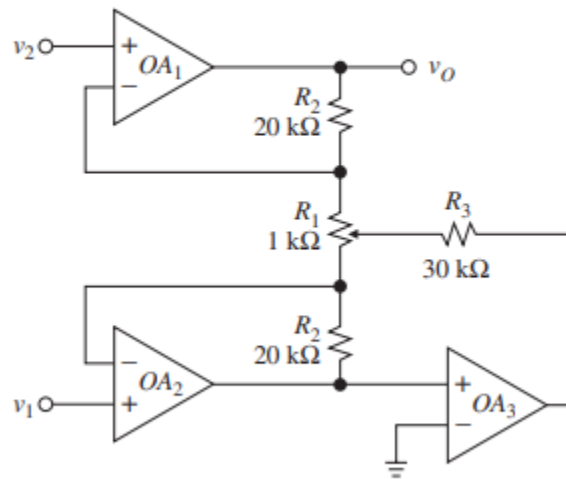


FIGURE P2.43

for a stable circuit, $V_{O2} = 0$, $V(-OA2) = V1$, $V(-OA1) = V2$

$$V_O - V2 = iR_{2.1} * R2$$

$$V2 - V_{pot} = iR_{1.1} * R1 / 2 = iR_{2.1} * R1 / 2$$

$$V_{pot} - V1 = iR_{1.2} * R1 / 2 = iR_{2.2} * R1 / 2$$

$$V1 - 0 = iR_{2.2} * R2$$

$$iR_{2.1} = (V_O - V2)/R2$$

$$iR_{2.2} = V1/R2$$

$$V_{pot} = V2 - iR_{2.1} * R1 / 2$$

$$V_{pot} = V1 + iR_{2.2} * R1 / 2$$

$$V2 - iR_{2.1} * R1 / 2 = V1 + iR_{2.2} * R1 / 2$$

$$V2 - (V_O - V2)/R2 * R1 / 2 = V1 + V1/R2 * R1 / 2$$

$$V2 - V_O/R2 * R1 / 2 + V2/R2 * R1 / 2 = V1 + V1/R2 * R1 / 2$$

$$V2 - V_O R1/2R2 + V2 R1/2R2 = V1 + V1 R1/2R2$$

$$V_O (-R1/2R2) + V2 (1+R1/2R2) = V1 (1+R1/2R2)$$

$$V_O (-R1/2R2) = V1 (1+R1/2R2) - V2 (1+R1/2R2)$$

$$V_O (-R1/2R2) = (V1 - V2) (1+R1/2R2)$$

$$V_O = (V1 - V2) (1+R1/2R2) (-2R2/R1)$$

$$V_O = (V1 - V2) ((-2R2/R1) + (-2R2/R1)R1/2R2)$$

$$V_O = (V1 - V2) (-2R2/R1 - 1)$$

$$V_O = (1 + 2R2/R1) (V2 - V1) \text{ QED}$$

P2) Book Problem 2.47: Assume low-frequency operation.

2.47 The dual-op-amp IA of Fig. P2.47 (see *Signals and Noise, EDN*, May 29, 1986) offers the advantage that by proper adjustment of the pot, a fairly high CMRR can be achieved and maintained well into the kilohertz range. Show that $v_O = (1 + R_2/R_1)(v_2 - v_1)$.

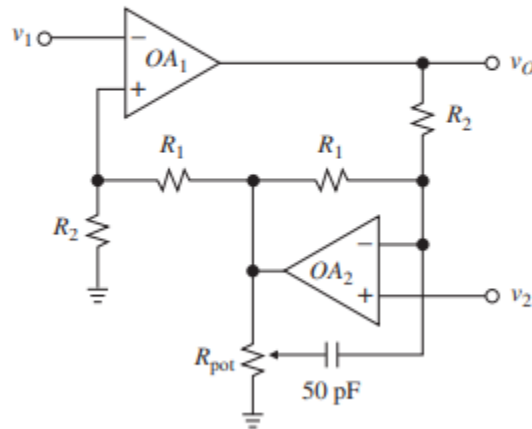


FIGURE P2.47

low frequency: cap is open circuit

Rpot is just from Vop2 to gnd, has no effect elsewhere

for steady state: $V(+OA1)=V_1$, $V(-OA2)=V_2$

$$V_1 = V_n (R_2/(R_1+R_2))$$

$$V_2 = (V_o - V_n) (R_1/(R_1+R_2)) + V_n$$

$$V_n = V_1 (1 + R_1/R_2)$$

$$V_2 = (V_o - V_1 (1 + R_1/R_2)) (R_1/(R_1+R_2)) + V_1 (1 + R_1/R_2)$$

$$V_2 = (V_o - V_1 ((R_1+R_2)/R_2)) (R_1/(R_1+R_2)) + V_1 (1 + R_1/R_2)$$

$$V_2 = (V_o (R_1/(R_1+R_2)) - V_1 (R_1/(R_1+R_2)) ((R_1+R_2)/R_2)) + V_1 (1 + R_1/R_2)$$

$$V_2 = V_o (R_1/(R_1+R_2)) - V_1 (R_1/R_2) + V_1 (1 + R_1/R_2)$$

$$V_2 = V_o (R_1/(R_1+R_2)) + V_1$$

$$V_2 - V_1 = V_o (R_1/(R_1+R_2))$$

$$V_o = (V_2 - V_1) / (R_1/(R_1+R_2))$$

$$V_o = (V_2 - V_1)(R_1+R_2)/R_1$$

$$V_o = (1 + R_2/R_1)(V_2 - V_1) \text{ QED}$$

P3) Book Problem 2.58: Assume a $1000\ \Omega$ Pt RTD (example resistance-vs-temperature table <https://www.sterlingsensors.co.uk/pt1000-resistance-table>). A Pt1000 RTD has a sensitivity of approximately $0.00385\ \Omega/\Omega/^\circ\text{C}$ (DIN 43760 Standard). Also assume that this is for a low-power implementation (hence the PT1000 RTDs), therefore target a current of $0.1\ \text{mA}$ through each RTD. How much power is being consumed by all the resistors in the circuit?

2.58 Assuming that $V_{\text{REF}} = 2.5\ \text{V}$ in Fig. 2.39, specify suitable component values for an output sensitivity of $0.1\ \text{V}/^\circ\text{C}$ with a Pt RTD.

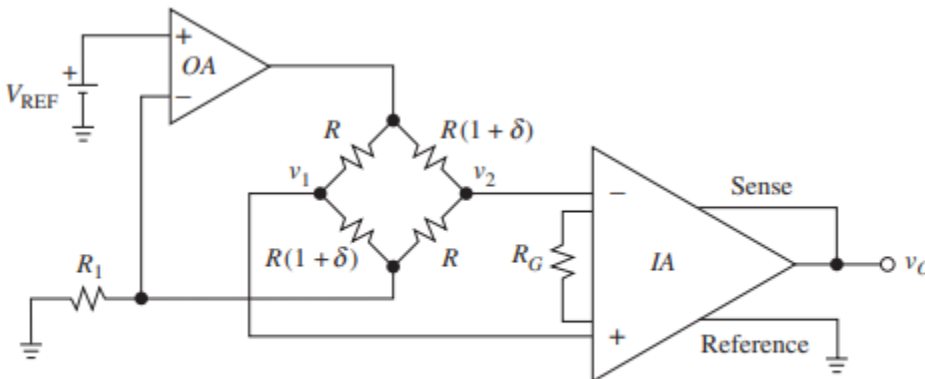


FIGURE 2.39

Bridge linearization by constant-current drive.

$0.1\text{V}/^\circ\text{C}$ at output

$$I_s = 2 \times 0.1\text{mA} = 0.2\text{mA}$$

$$I_s = I_{R1} = V(-OA)/R1 = V_{\text{ref}}/R1$$

$$0.2\text{mA} = 2.5/R1$$

$$R1 = 2.5 / 0.2\text{m} = 12500\Omega$$

$$V_o = R_G \Delta I_s / 2 = 0.1 \Delta = A (0.1 \Delta)$$

$$0.1\text{V}/^\circ\text{C} = A (0.1 \Delta) = A (0.1 \times 0.00385)$$

$$A = 259.74025974$$

$$R_G \cdot 0.2\text{m} / 2 = 0.1$$

$$R_G = 1\text{k}$$

$$P = 2.5 \times 0.2\text{m} + 4(0.1\text{m}^2 \times 1\text{k}) = 0.5\text{m} + 4(0.01\text{m}) = 500\mu + 4(10\mu) = 540\mu\text{W}$$

$$P = 540\mu\text{W}$$